

Intro

stability of plasma systems

Fluid stability $\rightarrow$ stable after kinetic effects

2-stream instability

Mathematical perspective:

PDE - constrained optimization for Vlasov - Poisson

Strang Splitting

e.g. VP: $f_t + v f_x + E f_v = 0$

A: $\partial_t f + v \partial_x f = 0$ speed

B: $\partial_t f + E \partial_v f = 0$ accelerate

1 time step: $f^{(1)} = S_A(\Delta t/2) f^n$
 $f^{(2)} = S_B(\Delta t) f^{(1)}$
 $f^{n+1} = S_A(\Delta t/2) f^{(2)}$

Sec 1 Vlasov - Poisson (with E)

$$\begin{cases} \partial_t f + v \cdot \nabla_x f - E_f^t \cdot \nabla_v f = 0 \\ E_f^t = H + \nabla_x \cdot V f \\ \Delta V f = 1 - \rho_f(t, x) = 1 - \int f dv \end{cases}$$

characteristic line: $\dot{x} = v$ $\dot{v} = -E^t = -(H + \nabla V f)$

Initial condition: $f_0(x, v)$

Sec 2 Constrained - optimization.

$$J(f|H) = \frac{1}{2} \|f|H(t)\|_{L^2(x,v)}^2 - f^{eq} \|_{L^2(x,v)}^2$$

Note: Green Function

$$\min_H J(f|H)$$

$$Lu = S$$

$$\text{s.t.} \begin{cases} \partial_t f + v \partial_x f - (H + E|f|) \partial_v f = 0 \\ E|f| = \partial_x G * (1 - \rho_f) \\ f|_{0,x,v} = f_0 = f^{eq} + \tilde{f} \end{cases}$$

Intro $G: L_x G(x, y) = \delta_y(x) \rightarrow \int_{\mathbb{R}} \delta_y(x) \phi(x) dx = \phi(y)$ for $\forall$ smooth ϕ

if $Lu = S$ then $u(x) = \int_{\mathbb{R}} G(x, y) S(y) dy$

$$Lu(x) = L_x \int_{\mathbb{R}} G(x, y) S(y) dy$$

$$= \int_{\mathbb{R}} L_x G(x, y) S(y) dy$$

$$= \int_{\mathbb{R}} \delta_y(x) S(y) dy$$

$$= S(x)$$

$$G: LG = \partial_{xx} G = \delta_0.$$

Lagrangian multiplier

$$\text{Cal } \frac{dJ}{dH}$$

$$L(f, H, g, \eta) = J(f)$$

$$- \langle \partial_t f + v \partial_x f - (H + \partial_x G * (1 - \rho_f)) \partial_v f, g \rangle_{x,v,t}$$

$$- \langle f|_{0,t} - f_0, \eta \rangle_{x,v}$$

$$\Rightarrow \min_{f, H, g, \eta} L(f, H, g, \eta)$$

$$\frac{\partial L}{\partial f} = \frac{\partial H}{\partial f} + \left(\frac{\partial H}{\partial f} \right) \frac{\partial L}{\partial f}$$

$$\frac{dJ}{dH} = \frac{\partial H}{\partial f} + \left(\frac{\partial H}{\partial f} \right) \frac{\partial L}{\partial f} \rightarrow \text{hard to calculate.}$$

$$\partial_H L = \langle \partial_v f, g \rangle_{x,v,t}$$

$$\text{set } \partial_f L = 0. \text{ (choose } g) \Rightarrow \frac{dJ}{dH} = \partial_H L = \langle \partial_v f, g \rangle_{x,v,t}$$

$$(2.5) \quad L(f, H, g, \eta) = J(f) - \langle f|_{0,t}, g|_{0,t} \rangle_{x,v} + \langle f|_{0,t}, g|_{0,t} \rangle_{x,v}$$

$$+ \langle f, \partial_t g + v \partial_x g - H \partial_v g \rangle_{x,v,t} - \langle f|_{0,t} - f_0, \eta \rangle_{x,v} - \langle \partial_x G * (1 - \rho_f) f, \partial_v g \rangle_{x,v,t}.$$

$$\langle (\partial_x G * \rho_f) f, \partial_v g \rangle_{x,v,t} = \langle f(t, x, v), \langle \partial_x G(x-y) f(t, y, w) \partial_v g(t, y, w) \rangle_{y,w} \rangle_{x,v,t}.$$

$$\frac{\partial L}{\partial f}(t, x, v) = 0, \quad 0 \leq t < T \quad \frac{\partial L}{\partial f}(T, x, v) = 0.$$

平移不变性: $G(x, y) = G(x-y)$

$$\Rightarrow u(x) = \int_{\mathbb{R}} G(x-y) S(y) dy = G * S$$

$$\Rightarrow \partial_t q + v \partial_x q - H \partial_v q + L \partial_x q^* (p_f - 1) \partial_v q + \partial_x q^* \langle f \partial_v q \rangle = - \frac{\partial J}{\partial f}(t, x, v).$$

$$g(t, x, v) = \frac{\partial J}{\partial f(t)}(x, v) = f(t) - f^{eq}$$

i.e. f - state variable

g - adjoint variable

$\Rightarrow$ Adjoint method:

$$\frac{\delta J}{\delta H}(x) = \int_0^T \int_{\mathbb{R}} \partial_v f(t, x, v) g(t, x, v) dv dt$$

$$f: \begin{cases} \partial_t f + v \partial_x f - (H + E f) \partial_v f = 0 \\ f(0, x, v) = f_0(x, v) \end{cases}$$

$$g: \begin{cases} \partial_t g = 0 \quad (\text{adjoint equation}) \\ g(T, x, v) = f(T, x, v) - f^{eq}(x, v) \end{cases}$$

$$H \rightarrow f[1H] \mapsto J(f[1H])$$

$$\frac{dJ}{dH} = \frac{d}{dH} J(f[1H])$$

$$\partial_H J(f) |_{\text{fixed } f} = 0.$$

$$\frac{dJ}{dH} = \partial_H L + (\partial_H f) \partial_f L \stackrel{\text{set } = 0}{=}$$

Gradient Descending:

1. for (2.2) : $f_0, f^{eq}, \text{TOL}, h, H^0$

$\rightarrow$ for $f[1H^n]$ $\leftarrow$ for $g[1H^n]$

compute $\frac{dJ}{dH}|_{H^n} = \langle \partial_v f[1H^n], g[1H^n] \rangle_{v,t}$

$$H^{n+1} = H^n - h \frac{dJ}{dH}|_{H^n}$$

Separately design calculator for $\rightarrow$ and $\leftarrow$
 $\hookrightarrow$ incompatibility

Sec 3 Discretize then Optimize

3.1 Discretization for VP

Semi-Lagrangian Track back the characteristics & perform the interpolation/projection

(Strang Splitting, Hamilton structure)

$$\partial_t f + v \partial_x f = 0$$

$$\partial_x f - (E + H) \partial_v f = 0.$$

$$\Rightarrow f^{n+1} = e^{\frac{\Delta t}{2} A} e^{\Delta t B(E^{n+1/2})} e^{\frac{\Delta t}{2} A} f^n$$

$$E^{n+1/2} = E \mathbb{I} e^{\frac{\Delta t}{2} A} f^n$$

$$\text{i.e. } f^*(x, v) = f(t_n, x - \frac{v \Delta t}{2}, v)$$

$$\partial_x E^{n+1/2} = \mathbb{I} - P_f$$

$$f^{**}(x, v) = f^*(x, v + (E^{n+1/2} + H) \Delta t)$$

$$f(t_{n+1}, x, v) = f^{**}(x - \frac{v \Delta t}{2}, v)$$

Remark: error estimate for $E^{n+1/2}$ (computed by f^*)

$$P_f^* = P_f(t_{n+1/2}) + O(\Delta t^2)$$

(Lie Splitting)

$$f^{n+1} = e^{\Delta t B(E^0)} e^{\Delta t A} f^n$$

$$(e^{\frac{\Delta t}{2} A} e^{\Delta t B(E^{n+1/2})} e^{\frac{\Delta t}{2} A})^* = e^{-\frac{\Delta t}{2} A} (e^{\Delta t B(E^{n+1/2})})^* e^{-\frac{\Delta t}{2} A}$$

$$\text{i.e. } g^{n,*} = e^{-\frac{\Delta t}{2} A} g^n. \quad g^{n,*} = (e^{\Delta t B(E^{n+1/2})})^* g^{n,*}. \quad g^{n+1} = e^{-\frac{\Delta t}{2} A} g^{n,*}$$

On space $C_{ij} = [x_{i-1/2}, x_{i+1/2}] \times [v_{j-1/2}, v_{j+1/2}]$

$$f_{ij}^n \rightarrow f^n(x, v) \text{ in } C_{ij}$$

$$\Rightarrow f^n(x, v) \approx \sum_{ij} f_{ij}^n \chi_{C_{ij}}(x, v)$$

In fully discrete setting.

$$e^{\frac{\Delta t}{2} A} f^n(x, v) = f^n(x - \frac{v \Delta t}{2}, v) = \sum_{ij} f_{ij}^n \chi_{C_{ij}}(x - \frac{v \Delta t}{2}, v)$$

no mesh $\Rightarrow$ Interpolation

$$-\frac{v \Delta t}{2 \Delta x} = n(v) + \alpha(v) := \lfloor -\frac{v \Delta t}{2 \Delta x} \rfloor + (-\frac{v \Delta t}{2 \Delta x} - \lfloor -\frac{v \Delta t}{2 \Delta x} \rfloor)$$

$$\Rightarrow f_{ij}^{n,*} = (1 - \alpha(v_j)) f_{i+n(v_j), j}^n + \alpha(v_j) f_{i+n(v_j)+1, j}^n \quad (\text{Linear})$$

Translate the formula into a matrix form

$$f_{:,j}^{n,*} = A^v_j f_{:,j}^n, \quad \text{where } A^v_{kl} = \begin{cases} 1 - \alpha(v) & l = k + n(v) \\ \alpha(v) & l = k + n(v) + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(A^v)^T = A^{-v}$$

then compute $E^{n+1/2}(x)$

$$E^{n,*} \approx E^{n+1/2}, \quad E^{n,*} = \mathcal{I} E_i^{n,*} = \mathcal{I} f^{n,*}$$

$$E: \mathbb{R}^{n_x \times n_v} \rightarrow \mathbb{R}^{n_x}$$

$$E(f) = C(1 - p_f), \quad C_{ij} = \partial_x G(x_i - x_j) \quad (p_f)_i = \frac{1}{n_v} \sum_j f_{ij}$$

$$\text{FFT in practice} \quad g(x) = 1 - p_f(x)$$

$$E = \partial_x V \quad \partial_{xx} V = g \quad (\partial_x E = g)$$

$$\Rightarrow \mathcal{F}(\partial_x E)_k = -ik \hat{E}_k$$

$$\Rightarrow -ik \hat{E}_k = \hat{g}_k, \quad \hat{E}_k = \frac{i}{k} \hat{g}_k$$

$$\Rightarrow E(f) = \mathcal{F}^{-1} \mathcal{I} \text{diag}(0, \frac{i}{1}, \dots, \frac{i}{n_x-1}) \mathcal{F}(1 - p_f)$$

B same as A.

$\Rightarrow$ conclusion of semi-Lagrangian

reformulation (discretization)

$$\min_H J(f, H) = \frac{1}{2} \sum_{ij} |f_{ij}^N - f_{ij}^{eq}|^2 \Delta x \Delta v$$

$$\vec{f} := [f^0, f^1, \dots, f^N]$$

$$L(H, f, f^*, f^{**}, g, g^*, g^{**}) = J(f) + \text{Term I} + \text{Term II} + \text{Term III}$$

$$L_R^n = -f^{n,*} + A f^n = 0 \quad R_2^n = -f^{n,**} + B(E(f^{n,*}) + H) f^{n,*} = 0 \quad R_3^n = -f^{n+1} + A f^{n,**} = 0$$

$$\text{Term I} = \Delta x \Delta v \sum_{n=0}^{N-1} \sum_{ij} [1 - f_{ij}^{n,*} + (1 - \alpha(v_j)) f_{i+n(v_j), j}^n + \alpha(v_j) f_{i+n(v_j)+1, j}^n] g_{ij}^{n,*}$$

$$\text{Term II} = (\alpha \text{ v-step}) \Delta x \Delta v \sum_{n=0}^{N-1} \sum_{ij} [1 - f_{ij}^{n,**} + (1 - \alpha(a_i^n)) f_{i+j+n(a_i^n), j}^{n,*} + \alpha(a_i^n) f_{i+j+n(a_i^n)+1, j}^{n,*}] g_{ij}^{n,**} \quad (a_i^n = E_i^{n,*} + H_i)$$

$$\text{Term III} = (\text{half a x-step}) \Delta x \Delta v \sum_{n=0}^{N-1} \sum_{ij} [1 - f_{ij}^{n+1} + (1 - \alpha(v_j)) f_{i+n(v_j), j}^{n+1} + \alpha(v_j) f_{i+n(v_j)+1, j}^{n+1}] g_{ij}^{n+1}$$

$$\nabla_H J = \nabla_H L + (\nabla_H f) \nabla f L$$

$$\Rightarrow \nabla f L = 0 \quad \frac{\partial L}{\partial H_i} = \frac{1}{\Delta x \Delta v} \sum_{n=0}^{N-1} \sum_j (f_{i,j+m_i^{n,k}+1}^{n,*} - f_{i,j+m_i^{n,k}}^{n,*}) g_{ij}^{n,*}$$

$$1^\circ \quad (3.16) \quad 0 = \frac{\partial L}{\partial f_{ij}^N} = \Delta x \Delta v (f_{ij}^N - f_{ij}^{eq}) - \Delta x \Delta v g_{ij}^N$$

$$(3.17) \quad g^N = f^N - f^{eq} \quad (\frac{\partial L}{\partial f_{ij}^N})$$

$$2^\circ \quad g^{n+1} \rightarrow g^n$$

$$\text{RHS of Term III} = \sum_{ij} [1 - \alpha(v_j) f_{i+n(v_j), j}^{n,**} + \alpha(v_j) f_{i+n(v_j)+1, j}^{n,**}] g_{ij}^{n+1}$$

$$= \sum_{ij} f_{ij}^{n,*} [1 - \alpha(v_j) g_{i-n(v_j), j}^{n+1} + \alpha(v_j) g_{i-n(v_j)-1, j}^{n+1}]$$

$$\frac{\partial}{\partial f_{ij}^{n,*}} \Rightarrow g_{ij}^{n+1} = (1 - \alpha(v_j)) g_{i-n(v_j), j}^{n+1} + \alpha(v_j) g_{i-n(v_j)-1, j}^{n+1}$$

$$\text{i.e. } \tilde{g}_{:,j}^{n+1} = (A^v_j)^T g_{:,j}^{n+1} = A^{-v_j} g_{:,j}^{n+1}$$

$$(\text{Term II Similarly, } g_{ij}^n = (1 - \alpha(v_j)) g_{i-n(v_j), j}^{n,*} + \alpha(v_j) g_{i-n(v_j)-1, j}^{n,*})$$

$$\text{i.e. } g_{:,j}^n = (A^v_j)^T g_{:,j}^{n,*} = A^{-v_j} g_{:,j}^{n,*}$$

for term II, $RHS = \sum_{ij} f_{ij}^{n,*} [(1 - \beta_i^n) g_{ij-m_i^n}^{n,**} + \beta_i^n g_{ij-m_i^n-1}^{n,**}]$ ($a_i^n = E_i^{n,*} + H_i$, $m_i^n = n(2a_i^n)$, $\beta_i^n = \alpha(2a_i^n)$)

rmk: $E^{n,*} = E(f^{n,*})$

$$E(f + \epsilon \psi) = -C (P_f + \epsilon P_\psi) = E(f) + \epsilon E(\psi)$$

$$\Rightarrow \langle \nabla P E, \psi \rangle = \lim_{\epsilon \rightarrow 0} \frac{E(f^{n,*} + \epsilon \psi) - E(f^{n,*})}{\epsilon} = E(\psi)$$

$$\frac{1}{\Delta x \Delta v} \langle \nabla P L | f^{n,*}, \psi \rangle = - \langle g^{n,*}, \psi \rangle + \sum_{ij} \psi_{ij} [(1 - \beta_i^n) g_{ij-m_i^n}^{n,**} + \beta_i^n g_{ij-m_i^n-1}^{n,**}] + \frac{\Delta t}{\Delta x} \sum_{ij} f_{ij}^{n,*} E(\psi) [1 - g_{ij-m_i^n}^{n,**} + g_{ij-m_i^n-1}^{n,**}]$$

$$g_{ij}^{n,*} = (1 - \beta_i^n) g_{ij-m_i^n}^{n,**} + \beta_i^n g_{ij-m_i^n-1}^{n,**} + \frac{\Delta t}{\Delta v} E(\phi)_i \quad \phi_{ij} = f_{ij}^{n,*} (g_{ij-m_i^n}^{n,**} - g_{ij-m_i^n-1}^{n,**})$$

3° Gradient descending

$$\Delta_H J = \Delta_H L \quad H^{k+1} = H^k - h \nabla_H J|_{H^k}$$

Correspondence between method 1 & 3

Sec 4 Example 1: $f_{10,x,v} = \chi(x) \frac{\exp(-v^2/2)}{2\pi}$ $\chi(x) = \exp[-a(x-b)^2] \sin^2 \frac{x}{2}$

$$(x, v) \in [0, 4\pi] \times [-b, b] \quad a = 0.2, b = 2\pi, T = 20.$$

$$J(f) = \frac{1}{2} \| f(20, \cdot, \cdot) - f_{10}(\cdot, \cdot) \|_2^2$$

$$H(x) := \sum_{k \in K} a_k \sin(\frac{kx}{2}) \rightarrow (a_{k_1}, a_{k_2}, \dots)$$

rmk: optimization for a_k

$$H(x) = \sum_{k \in K} a_k \sin(\frac{kx}{2})$$

$$\frac{\partial J}{\partial a_k} = \int_X \frac{\delta J}{\delta H}(x) \frac{\partial H}{\partial a_k} dx$$

$$= \int_X \frac{\delta J}{\delta H}(x) \sin \frac{kx}{2}$$

$$K = 1, 2 \rightarrow \{1, 2\} \rightarrow \{1, \dots, 10\} \rightarrow \dots$$

Non-convex optimization $\xrightarrow{?}$ global opt strategy

differential evolution (with adjoint gradient).

np-it descending figure (F6)

Sec 5 Example 2: 2-stream instability

$$f_{10,x,v} = (1 + \alpha \cos(\beta x)) f^{eq}(v), (x, v) \in [0, 2\pi/\beta] \times [-b, b]$$

$$f^{eq}(v) = \frac{1}{2\sqrt{2\pi}} [\exp(-\frac{(v-\tilde{v})^2}{2}) + \exp(-\frac{(v+\tilde{v})^2}{2})]$$

$$\text{if } H=0, E \nearrow$$

$$J(f) = \frac{1}{2} \| f(40, \cdot, \cdot) - f^{eq}(\cdot, \cdot) \|_2^2$$

$$H(x) = \sum_{k \in K} a_k \cos(\frac{kx}{2}), K \subset \mathbb{N}$$

F9: parameter scanning for $K = \{1\} \{2\} \{3\} \{1, 2\}$

F10: two parameters

F11: 5 parameters (influenced by initial condition)